

46750 – Optimization in Modern Power Systems

Grading guidelines for the group assignments — criteria, levels, individual adjustment, and annotated examples

Scope. This document applies to all group assignments of the course. It defines the four criteria against which every question is graded (Section 1), the qualitative levels that decide how many of a question's points are awarded (Section 2), and how the group grade is turned into individual grades from the participation table (Section 3). The points available for each question, the deliverables and the weight of the assignment in the final grade are given in each assignment text. Sections 4 to 9 illustrate the criteria and levels with excerpts from reports of previous years.

1. The four grading criteria

Every question of an assignment is graded against the four criteria below. The points available for a question are split between the criteria according to what the question asks for (see the points table of the assignment); a question that requires no code or no numerical results carries no points under those criteria.

Mathematical models. The correctness, completeness and rigor of everything written in mathematical form: optimization problems, Lagrange functions, dual problems, KKT conditions, and the analytical conditions or closed-form solutions derived from them. Formulations are compact (sets, indices, parameters), every notation is introduced exactly once and then used consistently throughout the report, every new equation is explained individually, and from one question to the next only what changes is stated. Properties (convexity, strong duality, necessity and sufficiency of the KKT conditions) are stated correctly and justified.

Writing. The quality of the reasoning and of its exposition. Per question, this covers the answers to the discussion questions – motivation of modeling steps, interpretations "in your own words", qualitative analyses, pros and cons, recommendations – and whether the argument is clear, precise, complete and logically organized. A separate global allocation (5 points, graded once for the whole report) covers the overall structure and logical flow of the report, adherence to the template, formatting, readability of figures and tables, typographical and language errors, and the participation table.

Code. The submitted repository runs and reproduces the results reported, is organized in a modular way (data loading, model building, solving, plotting), is documented (README, docstrings, comments) so that the graders can run every experiment of the report with a single command, and extracts the primal and dual values that the report discusses. Errors in the mathematical formulation that are faithfully implemented are penalized under *Mathematical models*, not under *Code*.

Numerical results. The substance of the numerical work: results are correct, or at least consistent with the model and discussed critically when they are not; the experiments and scenarios are relevant and motivated; the figures and tables chosen are the right ones to support the point being made, and are complete (units, axes, legends); and the analysis is non-trivial, goes beyond describing the plots, and leads to concrete insights tied to the theory. The distinction with *Writing* is one of substance versus exposition: a correct and insightful analysis that is hard to follow loses points under *Writing*; a clearly written analysis that is trivial, too high-level or wrong loses points under *Numerical results*.

2. Levels

Table 1 describes, for each criterion, what distinguishes an *insufficient*, a *good* and an *excellent* answer. The points of a (sub-)question under a criterion are awarded in proportion to the level reached: roughly none of them for *insufficient*, most of them for *good*, all of them for *excellent*. Use it as a checklist of what to strive for; Sections 4–9 illustrate every level with excerpts from previous years' reports.

Criterion	Insufficient	Good	Excellent
Mathematical models	Formulations are missing, incorrect or inconsistent: undefined symbols, wrong signs, indices or units, constraints that do not express the text. Lagrange functions, duals or KKT conditions are wrong, or copied from the lectures without being adapted to the problem. Properties (convexity, strong duality, feasibility) are asserted without justification or wrongly.	All requested models, duals and KKT conditions are correct and complete; every symbol is defined once and used consistently; properties are stated with a justification. The derivations reach the requested closed forms and conditions, with at most minor gaps: a case not treated, a sign convention left implicit, a step skipped that the reader can fill in.	In addition: formulations are compact and general (sets, indices, parameters; only what changes from the previous question is restated); every case of a derivation is handled, including ties, active bounds and non-unique solutions; the assumptions on the data under which a result holds are made explicit; results are <i>proved</i> from the optimality conditions, not merely checked numerically.
Writing	Discussion questions are answered by restating the results or the definitions; interpretations are missing, vague or wrong; the argument cannot be followed, or theory and results are never connected. Globally: no clear structure, template not followed, many typos, unreadable figures or tables, participation table or AI disclosure missing.	Every discussion question is answered with a clear and correct argument in the group's own words; interpretations are tied to the model (which constraint, which dual variable, which parameter); each question is logically organized and each modeling step is motivated. Some redundancy, hedging or imprecision remains. Globally: sound structure and readable figures, few typos.	In addition: the argument is precise and economical — every claim is supported by a derivation, a number or a figure, and nothing is superfluous; interpretations go beyond the question asked (limits of validity, what would change the conclusion, practical implications); terminology is consistent throughout and the report reads as one narrative, with cross-references between questions instead of repetition. Globally: polished presentation.
Code	The repository does not run, or does not reproduce the numbers in the report; a single monolithic script, results copied by hand, no README or documentation; dual variables not extracted from the solver.	The repository runs from the README with one command per experiment of the report and reproduces every reported number; the code is modular (data loading, model building, solving, plotting), documented (README, docstrings, comments) and extracts the primal and dual values that the report discusses.	In addition: one parametrized model (class or function) serves all the questions instead of copied variants; experiments, sweeps and scenarios are defined declaratively and their outputs (tables, figures) are saved automatically; sanity checks are built in (e.g. closed-form solutions compared with the solver); the repository is clean — clear naming, no dead code, consistent structure.
Numerical results	Results are missing, wrong, or inconsistent with the	Results are correct and complete; the requested base	In addition: experiments are designed to test a stated

	model without the group noticing; figures or tables lack units, axes or legends; the "analysis" describes the plots ("the load increases") without explaining them; experiments are run without a purpose.	cases, sweeps and scenarios are run; figures and tables are readable, complete and the right ones for the point made; the analysis explains the results through the model (regimes, active constraints, dual variables) and states whether they confirm the derivations.	hypothesis and isolate one effect at a time (thresholds identified, one parameter varied); anomalies are spotted and explained (ties, non-unique solutions, numerical tolerances) rather than glossed over; the insights are generalized to the modeling question — what the result implies for the choice of model — using the minimum number of figures needed.
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Table 1: Qualitative levels for each grading criterion.

3. From the group grade to the individual grade

The group grade of an assignment is adjusted individually according to each student's overall participation share s , as reported in the participation table of the report. With n students per group the fair share is $s_{\text{fair}} = 100 \% / n$; with four students per group, the fair share is $s_{\text{fair}} = 25 \%$. The individual grade is the group grade multiplied by the adjustment ratio $r(s)$ defined as follows (with 15 % replaced by $0.6 s_{\text{fair}}$ for other group sizes) and illustrated in Figure 1:

$$r(s) = \begin{cases} 0 & \text{if } s < 15 \% \\ s / s_{\text{fair}} & \text{if } 15 \% \leq s \leq 25 \% \\ 1 & \text{if } s > 25 \% \end{cases}$$

A student contributing less than 15 % therefore receives no credit for the assignment; between 15 % and the fair share the ratio increases linearly from 0.6 to 1; a contribution at or above the fair share earns the full group grade. The weight of each assignment in the final grade is stated in the assignment text.

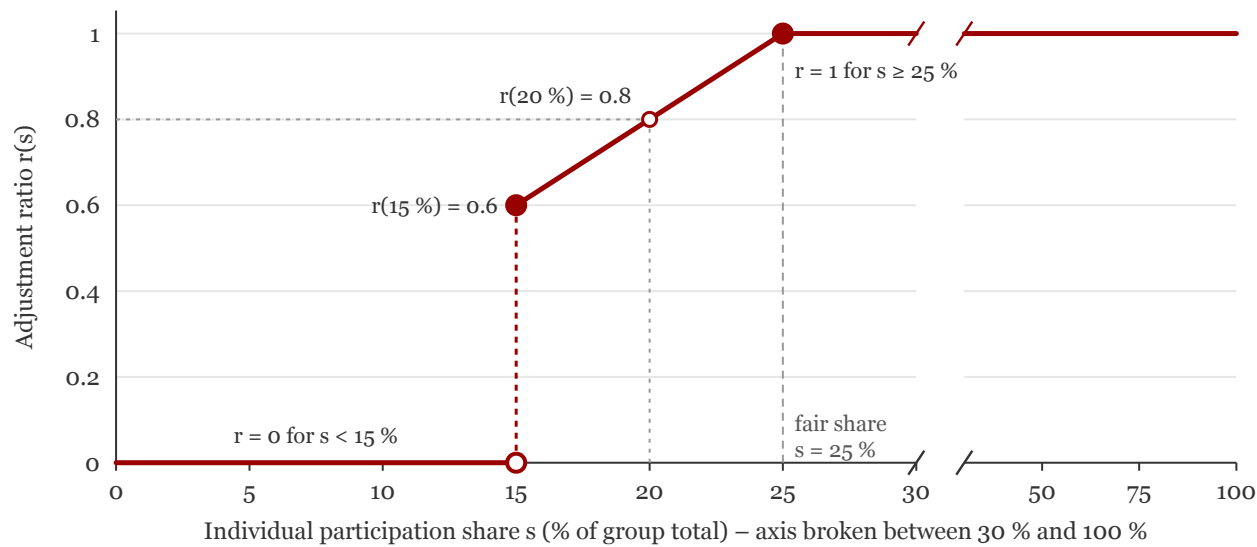


Figure 1: Individual grade adjustment ratio $r(s)$ as a function of the individual participation share s , for groups of four students (fair share 25 %). The ratio jumps from 0 to 0.6 at $s = 15 \%$ (open circle: value not attained; filled circle: attained value) and is constant above the fair share, so the horizontal axis is broken between 30 % and 100 %.

Annotated examples from previous years

How to read the examples. Sections 1 and 2 define the four grading criteria and, for each, what distinguishes an *insufficient*, a *good* and an *excellent* answer (Table 1). The sections below make those levels concrete with excerpts from real reports submitted in previous years of the course, together with the comments the graders wrote in the margin. Read it for the *skill* each excerpt shows, not for its content: the problems were different from this year's (a DC optimal power flow with a decomposition algorithm and battery placement in 2023; an open investment-planning project in 2024; a consumer-flexibility assignment in 2025, from which only excerpts that do not answer this year's questions are used). Every excerpt is quoted verbatim, shortened where marked [...], and anonymized; grader comments appear as **GRADER:** notes. Figures are reproduced as submitted.

Two things are worth knowing before reading. First, the reports labelled "good" here obtained high marks but still lost points — several of the *insufficient* examples below come from otherwise good reports, which is the most useful kind of example. Second, the graders' remarks recur from year to year: the same five or six mistakes cost most of the points lost. They are collected at the end.

4. Examples — Mathematical models

Insufficient	Good	Excellent
Formulations missing, incorrect or inconsistent (undefined symbols, wrong signs/indices/units, constraints that do not express the text); duals or KKT wrong or copied without adaptation; properties asserted without justification.	All requested models, duals and KKT conditions correct and complete; every symbol defined once and used consistently; properties justified; derivations reach the requested results with at most minor gaps.	In addition: compact and general (sets, indices, parameters; only what changes is restated); every case handled (ties, active bounds, non-uniqueness); assumptions on the data explicit; results proved from the optimality conditions, not only checked numerically.

Good and excellent

✓ Good — Define every set, index, parameter and variable before the model (2023, DC-OPF)

Let $\mathcal{G} = \{1, \dots, G\}$ denote the set of $G = 24$ generators, $\mathcal{W} = \{1, \dots, W\}$ the set of $W = 6$ wind farms, $\mathcal{D} = \{1, \dots, D\}$ the set of $D = 17$ demands, and $\mathcal{N} = \{1, \dots, N\}$ the set of $N = 24$ nodes. Furthermore, let $\mathcal{G}_i, \mathcal{W}_i, \mathcal{D}_i$ denote the subsets of generators, wind farms, and demands at node i . Importantly, $\mathcal{N}_i \subset \mathcal{N}$ denotes the subset of nodes connected to node i including node i itself. Now, let $p_i^{(G)}$ be the conventional production of generator $i \in \mathcal{G}$ and $c_i^{(G)}$ its cost of production. [...] Finally, let $\bar{P}_i^G$ and $\bar{P}_j^W$ denote the maximum production of generator $i \in \mathcal{G}$ and wind farm $j \in \mathcal{W}$, respectively. Now, the DC Optimal Power Flow (DC OPF) for a single hour can be formulated in the following way: [...] notice that decision variables are written in bold.

Why. Nothing is undefined when the reader reaches the model; node-dependent subsets ($\mathcal{G}_i, \mathcal{N}_i$) make the balance equation compact; every constraint that follows carries its quantifier (" $\forall i \in \mathcal{N}$ ") and the name of its dual variable after a colon. **GRADER:** "Very good!"

✓ Excellent — A nomenclature with units, and the set of decision variables written under "min"

(2023, model solution written by the teacher)

Sets and ranges: $\mathcal{G}$ set of all generators (synchronous and wind); $\mathcal{D}$ set of all loads; $\mathcal{N}$ set of all nodes; $\mathcal{G}_n$ set of all generators at node $n \in \mathcal{N}$ [...] Input parameters: $\bar{P}_g^G$ maximum capacity of generator $g \in \mathcal{G}$ (MW); [...] c_g production cost of generator $g \in \mathcal{G}$ (\$/MWh); VoLL value of lost load (\$/MWh); b_{nm} susceptance of line [...] The set of decision variables of the single-time step OPF $\Omega_{t_0}^P = \{p_{gt_0}\}_{g \in \mathcal{G}} \cup \{\theta_{nt_0}\}_{n \in \mathcal{N}} \cup \{sh_{dt_0}\}_{d \in \mathcal{D}}$ includes the production p_{gt_0} of each generator $g \in \mathcal{G}$ (including wind producers), the voltage angle θ_{nt_0} at each node $n \in \mathcal{N}$, and load shedding sh_{dt_0} for each load $d \in \mathcal{D}$. This single-time step OPF at time $t_0 \in \mathcal{T}$ is formulated as $\min_{\Omega_{t_0}^P} \sum_g c_g p_{gt_0} + \sum_d \text{VoLL} sh_{dt_0}$ s.t. [...]

Why. Parameters carry units; writing the decision-variable set under "min" removes any doubt about what is optimized and what is given; folding wind into $\mathcal{G}$ avoids duplicating every constraint. This is the standard the nomenclature section of your report template is meant to reach.

✓ **Good — Explain the dual problem, don't just write it** (2023, DC-OPF)

As evident from the dual formulation each primal constraint has resulted in one dual variable, while each primal variable has resulted in a corresponding dual constraint. Since $p_i^{(G)}$ and $p_i^{(W)}$ are non-negative, constraints (8)–(9) are of the form $\leq$. Meanwhile, constraints (10)–(11) are of the $=$ form, since θ_i is free. Furthermore, the coefficients of the dual variables in a given constraint in the dual formulation correspond to the coefficients of the primal variable [...] in all primal constraints, i.e., the matrix containing the coefficients is transposed.

Why. The primal–dual correspondence rules are stated explicitly (sign of a primal variable $\rightarrow$ type of the dual constraint; free variable $\rightarrow$ equality), which makes the dual checkable by the reader. **GRADER:** "Very good!"

✓ **Excellent — KKT conditions complete, compact and classified** (2024, bi-level investment planning)

1st order conditions: $c_{gt}^G - \lambda_t - \underline{\mu}_{gt} + \bar{\mu}_{gt} = 0 \quad \forall t, g$; [...] Complementarity conditions: $-p_{gt}^G \underline{\mu}_{gt} = 0$, $(p_{gt}^G - \bar{P}_g^G) \bar{\mu}_{gt} = 0$ [...]; Primal constraints: [...]; Dual constraints: $\underline{\mu}_{gt} \geq 0$, $\bar{\mu}_{gt} \geq 0$, $\lambda_t \in \mathbb{R}$ [...] (4a)–(4n).

Why. All four families are present (stationarity, primal feasibility, dual feasibility, complementarity), the lower bounds have their own multipliers, every line has its quantifier, and the block is short enough to be read at once. Compare with the insufficient examples below, each of which omits at least one family. (The teacher's 2023 model solution goes one step further and writes each inequality as $0 \leq \bar{\mu} \perp (p - \bar{P}) \leq 0$, carrying primal feasibility, dual feasibility and complementarity on one line.)

✓ **Excellent — Compactness by reference, and a one-sentence convexity statement** (2023, model solution)

$\min_{\Omega^{BESS}} \sum_t \sum_g c_g p_{gt} + \sum_t \sum_d \text{VoLL} sh_{dt}$ (12a) s.t. (11b)–(11l) (12b); $\sum_{n \in \mathcal{N}} u_{in} = 1, \forall i \in \mathcal{B}$ (12c) [...] where the set of decision variables $\Omega^{BESS} = \Omega^P \cup \{u_{in}\}_{i \in \mathcal{B}, n \in \mathcal{N}}$ includes all the decision variables Ω^P of the multi-time step OPF problem in Section 4 and the binary variables enforcing the placement of the batteries. [...] This minimization problem is non-convex, due to the binary variables u_{in} .

Why. Fourteen constraints are not re-typed; only what changes (one new constraint, the type of the new variables) is shown, and the reason for non-convexity is one sentence. This is exactly how Questions 2 and 3 of this year's assignment should build on Question 1: "the model of 1.(a) with ... replaced by ...".

✓ **Excellent — Deriving the dual as a method, not pasting it** (2025, consumer flexibility (generic paragraph only))

To evaluate the dual function, we minimize the Lagrangian over the nonnegative primal variables. Since the Lagrangian is linear in [the primal variables], this minimization acts independently on each variable. For a generic term αx with $x \geq 0$, we have $\min_{x \geq 0} \alpha x = 0$ if $\alpha \geq 0$ (minimum at $x = 0$), $-\infty$ if $\alpha < 0$ (problem unbounded below). Hence, the dual function is finite only if all coefficients of [the primal variables] in the Lagrangian are nonnegative. These nonnegativity requirements become the dual feasibility conditions. If they are satisfied, the minimizer simply sets all primal variables to zero, and the dual function reduces to the constant part of the Lagrangian.

Why. The reader sees the mechanism (regroup the Lagrangian by primal variable, read off the dual constraints, keep the constant part as the dual objective) and can apply it to any LP — including the ones in this assignment. The dual that followed in the report is not reproduced here.

✓ **Good — Motivating a modelling choice in words** (2025, battery question)

The end-of-day condition ($E_T = E_0$) is crucial for sustainable operation. It prevents the optimizer from achieving a superficially optimal result by simply depleting the battery's initial charge for a one-time profit. This ensures the battery ends the day with the same energy it started with, ready for the next cycle.

Why. A constraint that is not in the assignment text is introduced, and the sentence tells the reader what would go wrong without it. Another group modelled the same requirement as a soft constraint with a slack variable and a penalty, and explained which equations that changed — both are fine, because both are *argued*. Compare with the "implicit weight of 1 DKK/kWh" example below.

Insufficient

✗ **Insufficient — Constraints without quantifiers, sets and symbols never defined** (2023, DC-OPF)

$$\sum_{i \in G} M_{ni}^G p_i^G - \sum_{i \in L} M_{ni}^D p_i^D = \sum_{m \in N} B_{nm} \theta_m : \lambda_n (2); B_{nm}(\theta_n - \theta_m) \leq \bar{P}_{nm}^F : \mu_{nm}^F (3); \\ 0 \leq p_i^G \leq \bar{P}_i^G : \mu_i^G (4); \theta_1 = 0 : \sigma (5). [...] \text{ Lastly, the voltage magnitude at bus } m \text{ is given as } \theta_m.$$

Why. No " $\forall n \in \mathcal{N}$ ", " $\forall i \in \mathcal{G}$ ": the reader cannot tell how many constraints there are or what n, m, i range over; the set L is never defined, G_n appears later without definition, and θ is a voltage *angle*. **GRADER:** "Missing the running indexes. -0.5p"; "you do not define what all the variables and parameters are [...] -0.5p"; "What is the difference within set G_n and G ? -0.5p for clarity"; and, on the mapping matrices M^G, M^D : "A simpler/cleaner/more elegant way is to define a set that is dependent on n [...]. The model is then also only the size it needs to be." Later in the same report, when a " $\forall i \in T$ " finally appears: "THIS is what has been lacking. ALWAYS put this for ALL running indexes."

✗ **Insufficient — A section called "Dual formulation" that contains no dual problem** (2023, DC-OPF)

To derive the dual formulation of the problem, Lagrangian duality can be used. In order to derive the Lagrange function, it must first be shown that the objective function is convex. [...] The Hessian matrix is derived using Maple (see Appendix 6.1) and is seen to be equal to zero for all values. [...]

$$\frac{d}{dp_i^G} L = \sum_{i \in G} c_i^G + \sum_{i \in G} \mu_i - \sum_{i \in G} \mu_i + \sum_{i \in G} \lambda_i (17)$$

Why. The dual problem (objective and constraints) is never stated although the question asks for it; the derivative with respect to *one* p_i still contains sums over i ; three different multipliers are all called μ_i^G ; and a

Hessian computed in Maple to show that a *linear* function is convex is misplaced effort. **GRADER:** "The KKTs are good. But you do not explicitly write the dual problem? $-2.5p$ "; "What does the * indicate? $-0.5p$ ".

✗ Insufficient — KKT conditions announced, one family delivered (2025, consumer flexibility)

For the KKT conditions we take the primal feasibility + the dual feasibility + the complementary slackness. These conditions are necessary and sufficient for this convex LP: [followed only by complementary-slackness products of the form $\alpha_t (y_t - P_t^{pv}) = 0$]

Why. No stationarity conditions; the non-negativity constraints have no multipliers in the Lagrangian, so some of the "complementarity" products are in fact stationarity conditions of undeclared duals. This was the single most frequent remark in 2025: all three insufficient reports omitted the multipliers of the lower bounds. **GRADER:** "stationarity conditions missing"; "missing dual variables associated with the lower bounds on all variables in this Lagrange function (important for stationarity conditions)".

✗ Insufficient — Sign conventions that change between the dual and the Lagrangian (2025, consumer flexibility)

One report writes, in its dual problem, " $\mu_t \leq 0$ " for the multiplier of an upper bound $G_t \leq \bar{P}V_t$, then one page later, in its Lagrangian, " $\mu_t \geq 0$ the multipliers of $G_t - \bar{P}V_t \leq 0$ "; three of the four stationarity conditions that follow have signs that do not match the Lagrangian written just above them.

Why. Either convention is acceptable; changing it halfway makes every subsequent condition wrong. Derive the stationarity conditions *from the Lagrangian you wrote*, one derivative per variable, and check the signs against it. **GRADER:** "It seems you are confused on how to calculate the stationarity conditions. You need to look [at] the Lagrangian above and find the derivative with respect to every variable that you have [...] all the signs were messed up".

✗ Insufficient — Variables that are not defined, defined but unused, or "optional" (2025, consumer flexibility)

A report lists the decision variables L_t, I_t, X_t, C_t , then writes the balance as $L_t = G_t - X_t + I_t$ with a variable G_t that is never defined, never uses C_t , gives L_t no upper bound, and adds a PV cost term "(optional, if provided)" to the objective.

Why. The primal then has five variables and the dual four constraints, so they no longer correspond; the missing load bound lets the model buy the whole day's energy in one hour — which the report's own figure later shows, unnoticed. **GRADER:** "you are missing the definition of G here"; "You forgot to define an upper bound for your load consumption. This means that the model can import the daily demand in one hour."; "you now have 5 decision variables [...] Hence, the dual problem needs 5 constraints to match the primal problem and not 4 as you have."

✗ Insufficient — A wrong index on a dual variable (2023, multi-hour OPF)

$$[\text{nodal balance with batteries}] = \sum_m B_{nm} \theta_{mt} : \lambda_t \quad \forall i \in T \quad (64)$$

Why. The balance holds for every node *and* every hour, so its multiplier must carry both indices, λ_{nt} . In this year's assignment the same discipline applies to the hourly balance: one multiplier per hour, λ_t , never a single

λ. GRADER: "You are missing that this constraint (and the next) runs for all nodes. Hence, the LMP should have both the time and node index. -1p".

✗ Insufficient — Non-compact constraints and a wrong convexity argument (2023, battery placement)

$u_{ni}^B = 0 \vee u_{ni}^B = 1$ (85); $\sum_n u_{n1}^B = 1$ (86); $\sum_n u_{n2}^B = 1$ (87); $\sum_n u_{n3}^B = 1$ (88). [...] Since all constraints are linear i.e. there are no variables to the power of x , the hessian matrix will also be zero for the constraints and the conditions are true for this problem. This means that the optimal placement problem is convex, as both the objective function and the constraints are convex.

Why. Three identical constraints instead of " $\forall i \in \mathcal{B}$ "; a binary domain written as a disjunction instead of $u \in \{0, 1\}$; and a convexity argument that ignores the integrality of the feasible set. **GRADER:** "You could simply say that the sum of u over N must equal 1. -0.5p for compactness. You also do not explain these constraints. -1.5p"; "NO, due to the binary variables the problem is not convex. A problem can only be convex with a continuous feasible space. -2p".

✗ Insufficient — A misconception used to skip a derivation (2025, consumer flexibility)

In this exercise the dual, Lagrangian and KKT should have been derived. The important thing for this exercise is, that the quadratic term in the discomfort definition needs to be linearised. To avoid this two auxiliary variables will be introduced [...] This will be introduced in the code for part 1c.

Why. KKT conditions do not require linearity, and the question was left unanswered on the strength of a false premise. **GRADER:** "Why? You can derive the KKT conditions for any differentiable function, and they are necessary and sufficient for convex problems (under mild assumptions). In this case, [the] constraint is still convex."

✗ Insufficient, in an otherwise good report — Extending a model without defining the new parameters (2023, multi-hour OPF)

The battery model introduces SOC_i^{init} , ρ_i^{ch} , ρ_i^{dis} without defining them, quantifies the charging bounds with " $\forall i \in \mathcal{G}$ " instead of the set of batteries, drops the time index on $\theta_1 = 0$, and describes twelve new equations in one sentence: "The objective function remains the same as in the single step OPF, with the objective to minimize the cost of the electricity production subject to the balance equation, the update of the state-of-charge, [...]".

Why. The first model of the report had been exemplary (see the first good example); the rigour was not maintained when the model was extended, and it cost about seven points. This year's Questions 2 and 3 extend Question 1 — restate only what changes, but define and explain *all* of it. **GRADER:** "Should be defined, both the meaning and the value. -0.5p"; "-1.5p for not defining new parameters and not commenting on the addition of the time domain for all variables"; "Please describe each (new) equation explicitly like you do for (1)–(6). -1.5p"; "Time index?"

✗ Insufficient, in an otherwise excellent report — Notation drift (2025, consumer flexibility)

The multiplier of the daily energy requirement is μ on page 3 and ρ on page 7; α_t, β_t are the load-bound multipliers in one question and "the shadow prices on the positive and negative deviation variables" in the next; μ_{soc}^{up} and $\nu_{terminal}$ are used in the last question without ever being defined.

Why. Each slip is small; together they force the reader to re-derive what a symbol means in every section. Keep one nomenclature and add to it. **GRADER:** "have you used rho before? do you mean mu here?"; "for future work, you might want to use $\underline{}$ and $\overline{}$ and the same greek letter to define the duals for upper and lower bounds".

5. Examples — Writing

Insufficient	Good	Excellent
Discussion questions answered by restating results or definitions; interpretations missing, vague or wrong; the argument cannot be followed, or theory and results are never connected. Globally: no structure, template not followed, many typos, unreadable figures, participation table or AI disclosure missing.	Every discussion question answered with a clear, correct argument in the group's own words; interpretations tied to the model (which constraint, which dual, which parameter); each question logically organized, each modelling step motivated. Some redundancy or imprecision. Globally: sound structure, readable figures, few typos.	In addition: precise and economical — every claim supported by a derivation, a number or a figure, nothing superfluous; interpretations go beyond the question (limits of validity, what would change the conclusion, practical implications); consistent terminology, one narrative with cross-references instead of repetition. Globally: polished.

Good and excellent

✓ **Good — Constraint-by-constraint prose, including the modelling assumptions** (2023, DC-OPF)

The objective function (1) minimizes the total cost of power procurement. Constraint (2) models the power balance equation at each node with dual variable λ_i . Note that this is a simplification where real power losses and the reactive power balance are neglected, voltage magnitudes are assumed to be 1 p.u., and voltage angle differences are assumed to be small. Constraint (3) imposes the power flow limits of the cables between all nodes with dual variable μ_{ik} . Notice that this constraint is created in both directions meaning that effectively both an upper and lower limit for each line are imposed. [...] Constraint (6) fixes the voltage angle in node 1 to zero with dual variable ω . As only angle differences are included in the constraints an infinite amount of solutions exist — by setting $\theta_1 = 0$ one specific solution is chosen.

Why. Every equation number is referenced and its role explained; the assumptions behind the model are made explicit; the one non-obvious constraint (the reference angle) is justified. This is what "describe all new equations" means.

✓ **Excellent — Interpreting a dual variable: definition → meaning → numbers → structural cause** (2023, DC-OPF)

The locational marginal prices (LMPs) are the dual variables of the balancing constraints for each node, i.e., the increase in objective function if the demand at the node is increased by 1 MWh. Because the cheap generators are located in the left (and there is congestion on two lines connecting the left with the right), the LMPs are significantly lower in the left of the network with values in the range 5–10 \$/MWh while nodes in the right have values of 10–15 \$/MWh. The highest LMP is at node N14 which is at the end of a congested line.

Why. Four sentences: what the dual is, what it means economically, what values it takes, and *why* — traced to identified congested lines. This is the template for every "interpret the dual variable" item in this year's

assignment. **GRADER:** "Very good discussion! You show understanding of how the LMPs are effected by the network and congestion."

✓ **Good — Complementary slackness in words: binding constraint → non-zero dual → economics** (2023, DC-OPF)

Only G10 is fully dispatched and as a result only G10 has a dual variable different from zero as the power limit constraint associated with this variable is binding. If generator G10 could increase its capacity further, the objective value would decrease. It is also only generator G10 which is generating profits. [...] Thus, when generators G8, G9, and G11 have zero profits despite being active, it means that they are marginal generators.

Why. The chain binding $\Leftrightarrow$ non-zero multiplier $\Rightarrow$ what the multiplier would buy $\Rightarrow$ marginal versus infra-marginal is spelled out on the actual numbers. The same chain is what the curtailment and load-bound conditions of this year's Question 1 ask for.

✓ **Excellent — The two cases of a multiplier, spelled out** (2025, battery question)

The terminal shortfall multiplier θ corresponds to the end-of-day energy constraint $0.5E - e_{23} - s \leq 0$. It captures the terminal value of maintaining an adequate state of charge at the end of the day. [...] the complementary slackness condition $\theta(0.5E - e_{23} - s) = 0$ implies that: if the battery finishes the day with $e_{23} < 0.5E$, then the constraint is active and $\theta = w^{\text{depletion}}$, meaning the marginal value of one extra kWh of stored energy equals the penalty weight; if the battery ends with sufficient charge ($e_{23} > 0.5E$, hence $s = 0$), then $\theta = 0$, meaning no penalty is incurred.

Why. Both cases of the complementarity condition are translated into an economic statement. The grader's main remark on another excellent 2025 report was precisely that this was missing: "the only thing i am missing is more comments on the economic interpretation of the dual variables".

✓ **Good — Closing the loop between the qualitative analysis and the numbers, then concluding** (2025, battery question)

Comparison with Q1(c).iii. The numerical results confirm the qualitative expectations formulated earlier. The battery charges during PV or low-price hours and discharges during high-price periods, maximizing profit from price spreads. Greater flexibility (FLEX) increases export potential and battery throughput, leading to the highest profit, whereas comfort-oriented operation (RIGID) mirrors the BASE pattern with minimal additional value. [...] Conclusion. Introducing a battery transforms the consumer from a passive buyer into an active prosumer. Both primal and dual results demonstrate that storage enables temporal decoupling of consumption and generation [...]

Why. Every results section of this report ends with the same two paragraphs: "does this confirm what we derived?" and a two-sentence conclusion. The reader never has to reconstruct the link between theory and numbers. This year's Question 1.(g) asks for exactly this ("state whether the results confirm your analysis").

✓ **Excellent — Reading a figure quantitatively and turning it into the validity limit of an assumption** (2024, investment planning)

If the investment capacities are uncapped the model expects the NPV to increase linearly with the budget cap, as it will install cheap solar PV in node 24 where the capture prices are the highest with 55.99 €/MWh — expecting to get the same capture price no matter the installed capacity. In reality, starting from around a

budget of 500 M€, price cannibalization will be so severe that it will result in negative NPV [...] Figure 3 shows that only for small budgets of up to 250 M€ the new investments can approximately be expected to act [as] price takers in this scenario. After this the assumption no longer holds true.

Why. The mechanism is named, the figure is read with numbers, and the conclusion is a bounded, actionable statement about when a modelling assumption is valid — the kind of insight the "excellent" level asks for under both Writing and Numerical results.

✓ **Excellent — Limitations that say which way each assumption biases the result** (2025, battery investment)

First, the optimisation assumes perfect foresight and one PV production scheme, as well as one price scheme for the whole period. [...] Second, the battery is modeled with linear scaling and no degradation, implying constant efficiency and capacity over its lifetime. Including degradation would require multi-year state tracking and typically lead to smaller optimal capacities and longer payback periods. Third, the absence of discounting over the 10-year horizon slightly overestimates long-term returns, as future savings are valued equally to present ones.

Why. Not a list of "we could have done more", but a statement per assumption of what it does to the answer. Compare with the honest negative result below.

✓ **Good — Reporting a negative result with evidence instead of hiding it** (2024, investment planning)

A central aspect here, is that in both cases the consensus ADMM does not end up at the actual solution, but fails to dispatch the demands and generators correctly. [...] Crucially, convergence is only observed for the primal and dual residuals, but not for the consensus residual, which also underlines a gap in the implemented algorithm. Various values of the penalty parameter, γ , [were] tested, with values ranging from 1 to 500, but none of them alleviated the core issues described.

Why. The failure is stated, diagnosed (which residual does not converge) and bounded (what was tried). A wrong result analysed this way earns more than a wrong result presented as right — see the "sub-one-year payback" example below.

Insufficient

✗ **Insufficient — Sentences the reader cannot parse** (2023, decomposition)

When the two equation are relaxed each variable which is not linked to other sets of variables should get a sub-problem. These two equations links the nodes, therefore when relaxed the problem can be decomposed into 24 sub-problems, one for each node. Hence, the problem is decomposable, with two complicating constraints, and when these are fixed the problem can be decomposed into 24 sub-problems.

Why. The first sentence has no clear meaning, the conclusion is stated twice, and "relaxed" and "fixed" are used as synonyms although they are different operations. **GRADER:** "I do not understand what you mean here, -0.25p".

✗ **Insufficient — Describing colours instead of numbers, and sending the reader to the appendix** (2023, otherwise good report)

By looking at the figures above, it can be seen by the color of the nodes, that the nodes in the plot with BESS installed (fig. 3), generally have a turquoise color, which indicates a higher LMP value, than the blue color the nodes have in fig. 2. [...] By looking at figs. 6 and 7, it is seen that most nodes have roughly the same color [...] This means that for the BESS system, the LMP values fluctuate less, introducing a more stable price to the system.

Why. "Turquoise versus blue" replaces a quantitative statement; the figures the argument rests on are in the appendix; and the conclusion is contradicted by the plots. **GRADER:** "Referring directly to the appendix in the text, -0.5p"; "Looking at figure 2 and 3, it looks as if the battery yields more fluctuating prices between the nodes.. could comment on why that is?"

✗ Insufficient — A trivial fact presented as a finding (2023, DC-OPF)

Table 2 shows that the total production sums to 1775.835, which is equal to the set demand of the first hour. Therefore, it is seen that the generator production covers the given demand for this hour and the optimization problem has given a feasible solution.

Why. The balance constraint guarantees this; a sentence that restates a constraint is not an analysis. Contrast with the "Only G10 is fully dispatched" paragraph above, written about the same kind of table.

✗ Insufficient — Interpretation contradicted by the group's own figure (2025, consumer flexibility)

Under Base, the optimizer shifts flexible load into cheaper hours and maximizes PV self-consumption when available.

Why. The table two pages earlier reports 0 % self-consumption for the base case, and the figure on the same page shows the entire daily energy imported in hour 0 and nothing afterwards (see the Numerical results section). Fluent, generic text that was written without looking at the numbers is the most common way to lose Writing points while also losing Results points. **GRADER:** "your model imports the whole daily demand in the first hour. Therefore, you do not observe any demand shift [...] You should be more careful when reading the input data".

✗ Insufficient — Meta-commentary instead of an answer; group-process remarks in a technical report (2025, consumer flexibility)

In this part of the assignment, it should have been evaluated how this model is different than the first one. The first model in part a had a more fixed load throughout the day, where for this model has a more flexible load. [...] The code has been implemented by [a group member] but not analysed, I will try to write some context for it here for further analysis.

Why. Saying what should have been done is not doing it, and the division of labour inside the group belongs in the participation table, not in the analysis. Three vague sentences also earned the remark that a linear discomfort constraint had made the "new" model equivalent to the old one — which the group had not noticed.

✗ Insufficient — A result that should have triggered a sanity check, presented as an insight (2025, battery investment)

In all cases, the payback time increases with higher capital expenditures and larger storage capacities, yet remains below one year for the most flexible consumer under low-capex conditions. This indicates a strong

incentive for early investment when battery prices are moderate.

Why. A home battery that pays for itself in three months is not an insight, it is a symptom; the group's code had the sign of the objective wrong and reported profits for a cost minimization. The "excellent" level under Numerical results is largely about noticing such things. **GRADER:** "it seems that you confused the costs with profits and you have defined in your code to earn profits when importing power".

✗ Insufficient — A chronological log of attempts instead of an analysis (2023, battery placement)

It was not possible to make the code run correctly for this problem, so different steps of the process will be discussed instead. [...] When this was run, it was seen that the optimal placement of the BESS systems were decided to be two at node 20 and one at node 8. The optimal value for this problem was seen to be 151652.76, however this is the same as seen in Section 4.1, when the BESS systems where located at nodes 5, 16 and 23. This seemed an odd result and therefore, it was chosen to change the nodes [...] When these constraints were added the model claimed the optimal positioning of the three BESS systems to be at node 1, however, it did not show the BESS systems as either charging or discharging [...] This is an interesting result, as it is the best result obtained thus far, although the code used to find this optimal placement prints a different result.

Why. Honesty about a failure is welcome (compare the 2024 ADMM example), but this is a diary with mutually inconsistent numbers and no diagnosis. One sanity check — fixing the placement in the new model and verifying that it reproduces the previous result — would have located the bug; the grader did it for them: "I think the issue lies in that you define the balance constraint (75) wrong".

6. Examples — Code

Insufficient	Good	Excellent
Does not run or does not reproduce the report; monolithic script; results copied by hand; no README; duals not extracted.	Runs from the README with one command per experiment and reproduces every reported number; modular (data, model, solve, plot), documented, extracts the primal and dual values discussed.	In addition: one parametrized model serves all questions; experiments defined declaratively with outputs saved automatically; built-in sanity checks (closed forms vs solver); clean repository.

The reports themselves rarely show code, so the evidence here is what the reports say about their implementation and what the graders found when running it. Note that *none* of the three best 2025 reports described its code beyond a link to the repository; the code criterion was graded from the repository alone. A short paragraph per question saying which file and function implement which model, and which command produces which figure, is cheap and makes the grader's job — and your grade — easier.

Good and excellent

✓ Good — An explicit model-to-function mapping (2023, DC-OPF and decomposition)

The single time step DC OPF is implemented in the python function runSingleStepOPF. [...] The subproblem for node i is shown below (implemented in the function runSubNode) [...] This update step is performed in the following unconstrained subproblem (implemented in the function runSubTheta_bar) [...] [Algorithm 1] is implemented in the python function runConsensusADMM.

Why. Each block of equations names the function that implements it, so the reader can open the code and find it; the decomposition is visibly modular (subproblem, global update, outer loop are separate functions).

✓ **Excellent — Pseudocode that cites equation numbers, all tuning parameters reported, and a validation against a reference solution** (2023, decomposition)

Algorithm 1: while $e > \epsilon$ and $v < \text{max_iter}$ do; for $i \in \mathcal{N}$ do $p_i^{(G)}, \theta_{ki} \leftarrow \text{eqs. (40)–(42)} \triangleright \text{run subproblem for each node; for } i \in \mathcal{N} \text{ do } \bar{\theta}_i \leftarrow \text{eq. (45); } \lambda \leftarrow \lambda + \gamma(\theta_{ki} - \bar{\theta}_k) \triangleright \text{update } \lambda; e \leftarrow \|\lambda^{(v+1)} - \lambda^{(v)}\| / \|\lambda^{(v)}\|. [...] \text{The parameters have been tuned to } \gamma = 10, \text{max_iter} = 10\,000, \text{ and } \epsilon = 10^{-7}. \text{ The model converges in 7391 iterations. [...] The objective value of the solution is 4754.47 \$ which differs from the true optimal solution by 0.04 \$. The same four generators are activated as in the optimal solution and they produce the same within a margin of 0.04 MW.}$

Why. Anyone can re-run this with the same γ, ϵ and iteration cap and check the same 7391 iterations; and the decomposed code is validated against the centralized solve on the objective *and* the primal variables. In this year's assignment the equivalent check is comparing the closed-form solutions you derive with what the solver returns, hour by hour.

✓ **Good — A tuning choice documented with a small experiment** (2025, battery question)

How $w^{\text{depletion}}$ is calculated: $w^{\text{depletion}} = k \cdot \sum_t \lambda_t / 24$, where λ_t is [the] electricity price in hour t [...] After testing the optimization problem with $k = 0.5, k = 0.8$ and $k = 1$, the lowest k value resulted in complete storage depletion at the end of the day. To avoid that, [the] k factor was accordingly set to an intermediate value $k = 0.8$.

Why. The parameter is derived from the data, the test is shown (a figure with the three cases accompanies it), the choice is justified — reproducible. The same report also states its solver and status: "All numerical experiments converged to optimality (Gurobi barrier method, < 0.01 seconds per solve)".

✓ **Excellent — Implementation improvements quantified** (2024, investment planning)

[Table 6, model run times in seconds for 1 / 180 / 365 days:] Bi-level nodal investment planning 7.44 / 84.62 / >36000; + variable bounds (deduction) 4.14 / 62.61 / -; + λ upper bounds (from market clearing) 1.57 / 8.56 / -; + NPV bounds (from multi-step) 1.42 / 6.73 / 118.26. [...] All runtimes presented are based on runs performed on an Intel Core i5-1035G1 CPU @ 1.00 GHz.

Why. Each engineering iteration on the implementation is named and its effect measured; hardware is stated. This is evidence of a tested, modular code base without a single line of code in the report.

Insufficient

✗ **Insufficient — "Available on GitHub", four times** (2025, consumer flexibility)

The implementation code of this optimization problem is available on our GitHub repository, as noted in the assignment.

Why. This is the complete answer to every "implement" item of the report. Nothing about structure, inputs, how to run, or which script produces which figure; nothing in the report can be reproduced from the report.

✗ **Insufficient — Hand-written subproblems for two nodes, and code that does not run** (2023, decomposition)

To implement the decomposed problem in Python, it was chosen to only look at two nodes, to show how it would be done. The nodes chosen are n_1 and n_7 and their sub-problems are: [...] Unfortunately, these sub-problems were not fully coded and the delivered code was an attempt at looking at the relationship between node 1 and 2, however, the code does not run properly and many of the node relationships are missing.

Why. Writing out node-specific problems by hand is the opposite of a set-indexed, modular implementation – and it did not run. **GRADER:** "–4p for non-working codes [...] –4p for no results."

✗ Insufficient – Code and report that disagree, results that cannot be identical (2025, consumer flexibility)

A report states a cost *minimization* while the code evidently *maximizes*; two scenarios with different discomfort weights (0.01 and 1) return bit-identical objectives; imports of 1.4×10^{-11} kWh are printed as results; and a quantity declared as a decision variable in the model is later swept as a parameter in the experiments.

Why. None of these was tested before being written up. **GRADER:** "These results cannot be identical... Your model implementation is most likely wrong"; "from what I can see here you probably have solved your model in maximization for the previous questions, but in your report you have it as minimization. You need to be aware and compliant of your code and report implementation"; "this is a variable in your model you can vary it, its decided from the solver."

✗ Insufficient, in an otherwise excellent report – Figures and text from different runs; plotting artefacts (2025, battery investment)

The text reports "flexible consumers achieve 20–25 % ROI at 1500 DKK/kWh" while the figures on the next page show an ROI of about zero for every consumer type; another report's bar labels contain a literal "\n" ("Save 7.16\nDKK/day"); a third uses a developer axis label, "Payback (years); NaN means no positive cashflow".

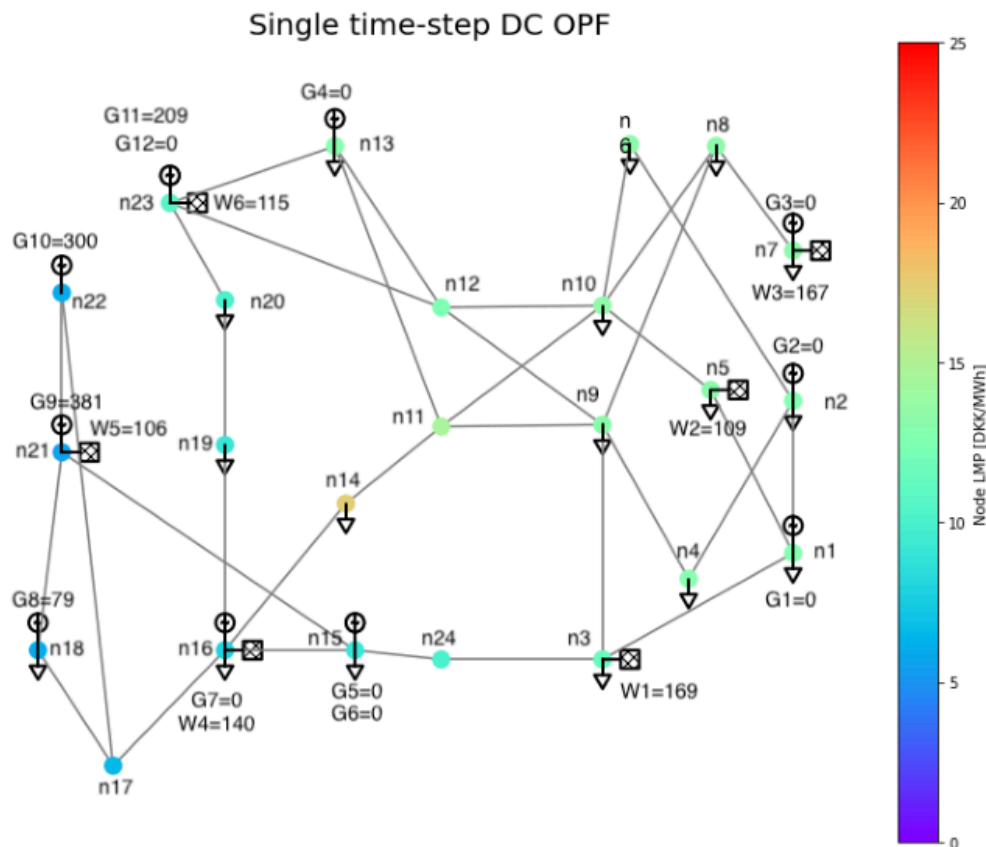
Why. When figures are not regenerated by the same script that produces the numbers in the text, they drift apart, and the grader cannot tell which is right. Save every figure and table from the code that computes them, and re-run everything once before submitting.

7. Examples – Numerical results

Insufficient	Good	Excellent
Results missing, wrong or inconsistent with the model without the group noticing; figures without units, axes or legends; "analysis" that describes the plots without explaining them; experiments without a purpose.	Results correct and complete; the requested base cases, sweeps and scenarios run; figures and tables readable, complete and the right ones; analysis explains the results through the model (regimes, active constraints, duals) and says whether they confirm the derivations.	In addition: experiments designed to test a stated hypothesis and isolate one effect at a time; anomalies spotted and explained; insights generalized to the modelling question, with the minimum number of figures.

Good and excellent

✓ **Excellent – One figure that answers the whole question** (2023, DC-OPF)

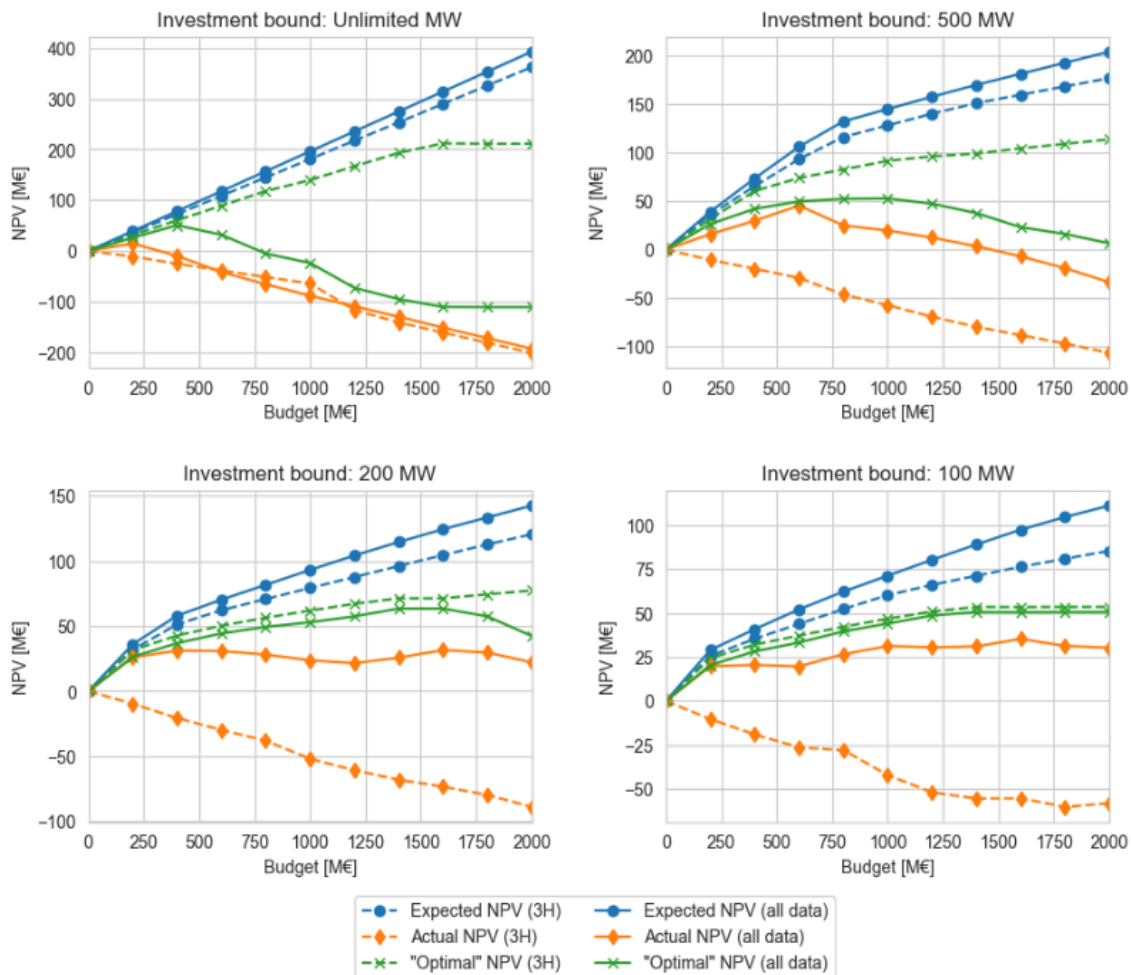


Original caption: "Network LMPs, generator production, and wind production for single time-step DC OPF. Generators are attached to their nodes as black circles, wind farms as squares, and loads as down-pointing arrows."

Why. Topology, prices (colour) and dispatch (labels) in one view, so the argument of the text — cheap generation on the left, congestion on two lines, the highest price at the end of a congested line — is visible; the caption explains the encoding. **GRADER:** "Very nice figure!" (Two small flaws the grader also noted: the colour bar says DKK/MWh while the text uses \$/MWh, and its range 0–25 wastes half the scale on values that do not occur.)

✓ **Excellent – A multi-panel figure that answers a research question along two dimensions**

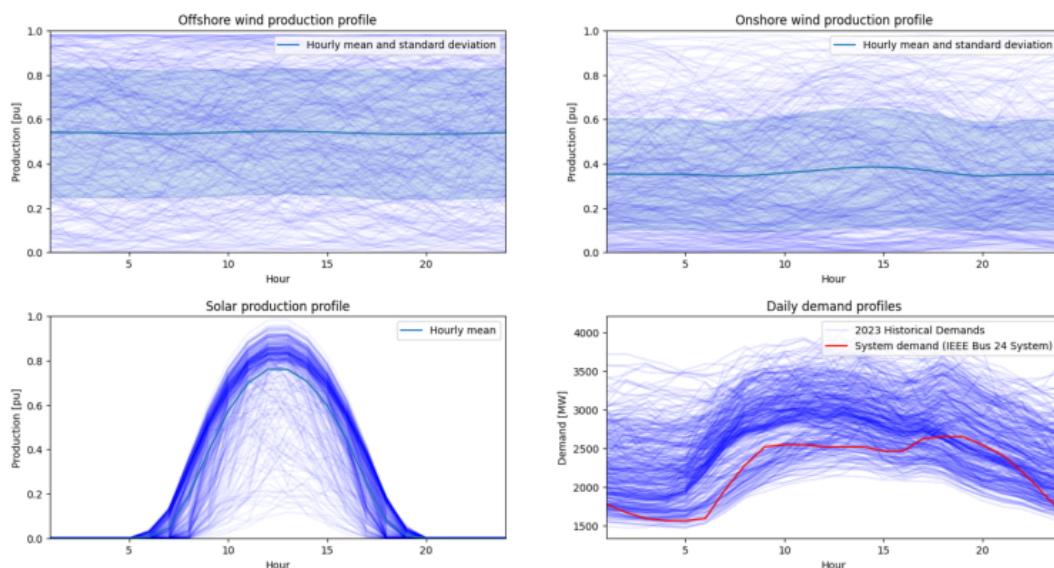
(2024, investment planning)



Original caption: "Expected and actual NPV from nodal investment together with the optimal NPV from the bi-level nodal investment. The dashed lines are evaluated on just three hours, while the solid on a full year."

Why. Consistent colours across panels, one shared legend, units on both axes, and a text that walks through each panel with numbers ("negative NPV for budgets of 800 M€ or more", "plateau at 1,600 M€ with an NPV of 200 M€"). The question "when is the price-taker assumption valid?" is answered by the figure itself.

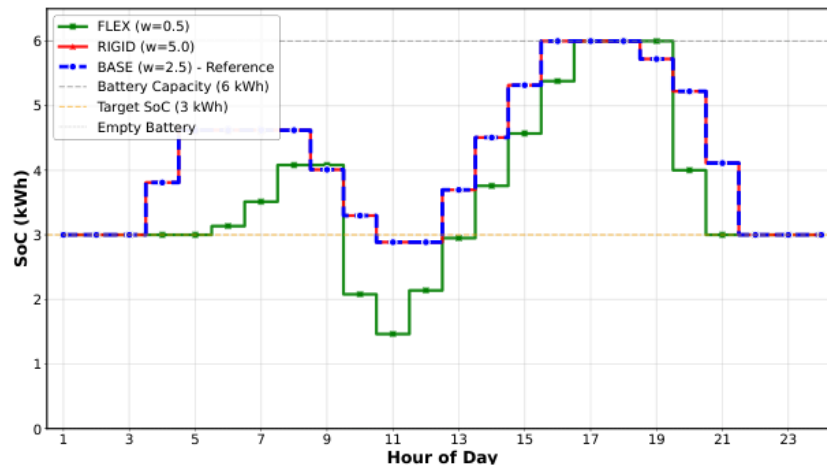
✓ **Good — Show the input data before the results** (2024, investment planning)



All 365 daily profiles drawn as translucent lines with the hourly mean; the demand panel overlays the reference IEEE profile in red. The caption states the data source, the weather year and the locations.

Why. The reader can later judge whether the results are plausible, and the comparison with the reference dataset justifies the group's claim that the data "require no further adjustment". In this year's assignment the equivalent is one figure of prices, tariffs and PV before Question 1's results.

✓ **Good — Reference lines that make a plot self-explanatory** (2025, battery question)



Original caption: "Daily battery state-of-charge for BASE, FLEX, and RIGID consumers." Horizontal lines mark the capacity, the target state of charge and the empty battery.

Why. The three horizontal reference lines let the reader see at a glance when a bound is active, which is what the text then discusses through the multipliers. Compare with the two unlabelled state-of-charge plots below.

✓ **Excellent — A with/without comparison quantified and tied to the duals** (2023, multi-hour OPF)

The objective value of the multi-step OPF without a BESS is 160 266.74 \$ while it drops to 155 607.54 \$ when a BESS is implemented. This is a difference of 4 659.2 \$ and a reduction of the cost by approximately 3%. [...] Without the BESS the congestion between node 14 and 16 occurs for 17 hours, with the BESS the congestion between these nodes occurs for 19 hours. [...] In total, congestion is happening for 27 hours without the BESS and 40 hours with the BESS. [...] Even though the congestion occurs for more hours when the BESS is implemented the dual variables associated with the maximum flow constraints are lower compared to when the BESS is not implemented. Consequently, the congestion has less influence on the objective value when the BESS is implemented and the gain which can be achieved by increasing the flow constraint limits is also lowered.

Why. Absolute and relative numbers; a non-obvious observation (more congested hours but lower congestion duals) is explained with the meaning of the dual, and the hourly duals are in an appendix table for verification.

✓ **Excellent — Experiments announced with a question, a motivation and metrics before any number** (2025, battery investment)

Experiment 1: Cost Sensitivity Analysis. We test battery capital costs of 800, 1500, and 2500 DKK/kWh across three consumer types — inflexible (high comfort weight), moderate, and flexible (low comfort weight [...]). This reveals cost thresholds for profitable investment by consumer segment. The motivation is straightforward: battery costs are declining rapidly, and identifying break-even points informs both

consumer decisions and manufacturer targets. Experiment 2: Price Volatility Impact. [...] Experiment 3: Tariff Structure Effects. [...] Methodology. We calculate profitability metrics by comparing scenarios: Net Profit [...] ROI [...] Payback Period [...] Optimal Capacity.

Why. One parameter per experiment, a stated reason, and the metrics fixed in advance — the design isolates one effect at a time, which is what the sweeps of this year's Questions 1.(g), 2 and 3 ask for. (The figures that followed in this report were too small to read — the grader's only complaint — which shows that design and execution are graded separately.)

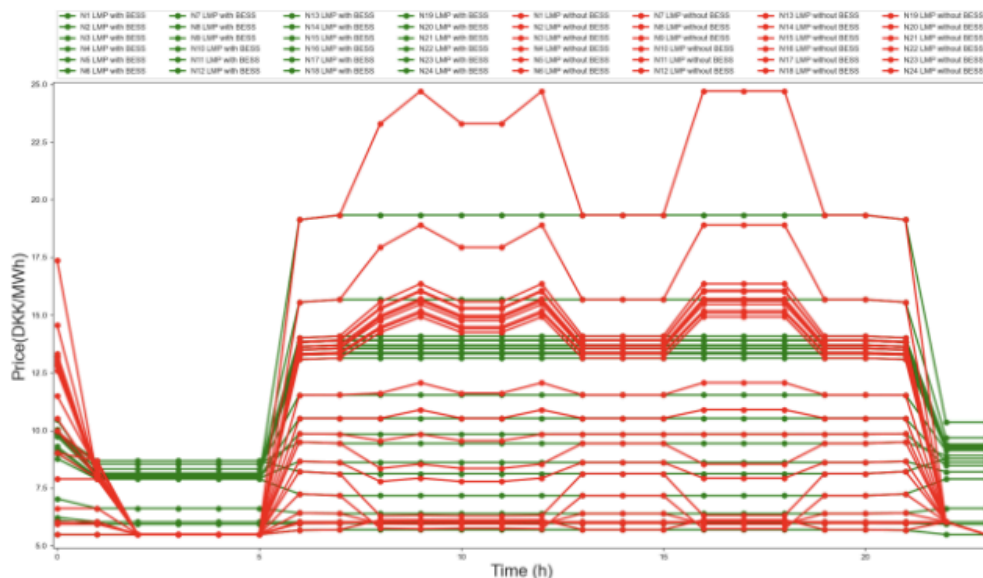
✓ **Good — Aggregate first, then justify where to zoom in** (2025, battery investment)

[Table: cost level, DKK/kWh, profitable scenarios, average optimal size, average payback:] Low 3000 — 5/9 — 5.7 kWh — 7.3 y; Baseline 6000 — 1/9 — 0.7 kWh — 9.0 y; High 8000 — 0/9 — 0.0 kWh — ∞. [...] Given that economic viability is observed almost exclusively at the lowest investment cost, the subsequent analysis and visualizations focus on the 3,000 DKK/kWh case.

Why. Nine scenarios summarized in three rows, and the decision of what to show in detail is taken from the numbers rather than by default. The same report anchors invented parameter values in a source ("Energinet has reported [...] willingness-to-pay values ranging from 22 to 174 DKK/kWh [...] substantially lower values are used to represent [...] routine load shifting").

Insufficient

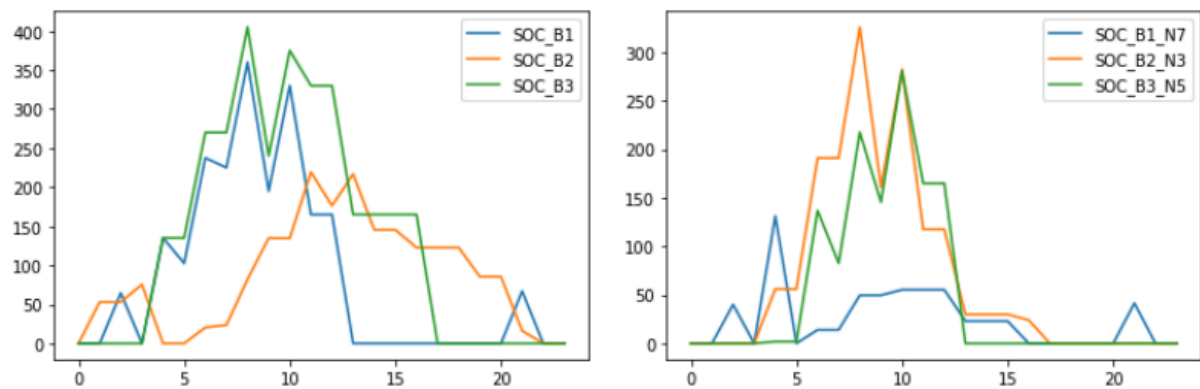
✗ **Insufficient — Forty-eight series in two colours** (2023, multi-hour OPF)



Original caption: "Dual variable associated with the balance constraint at each node, which is the LMP for the generators at the corresponding node. Red = without BESS, Green = with BESS."

Why. No individual node can be identified, so none of the text's claims ("in the hours of 3 to 5 and 23 to 24, the LMPs are cheaper without BESS") can be checked; the legend is unreadable and the y-axis unit (DKK) contradicts the tables (\$). A node × hour heat-map, a few representative nodes, or the *difference* between the two cases would each have carried the message. **GRADER:** "This is really a difficult plot to look at.... -1p in efficient illustration of results." The same report's next figure (24 generator cost curves, same two colours) earned "This is impossible to interpret. -2p".

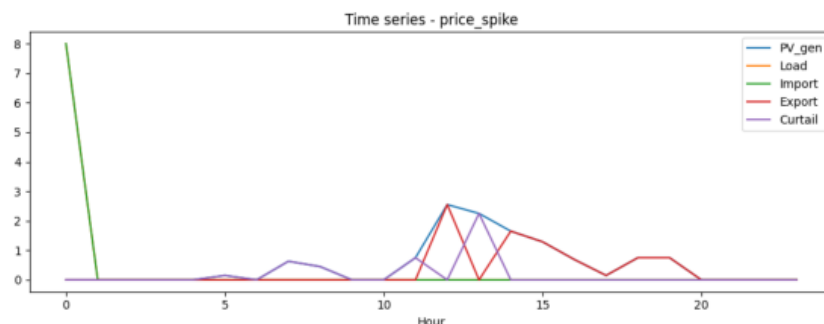
✗ Insufficient, in an otherwise good report — Side-by-side plots without labels and with different scales (2023, batteries)



Original captions: "TASK 3 – Battery SOC" and "TASK 4 – Battery SOC".

Why. No axis labels, no units, and two y-scales (0–400 and 0–300) for plots meant to be compared. **GRADER:** "Units and labels!! –0.5p"; "when you are comparing two plots side by side like this, it is nice if the scale on the y-axis remains the same as it gives a more intuitive understanding of the plots." The same report lost a further point for tables without units ("UNITS!! –0.5p"). Units are the cheapest points to lose in every year of this course.

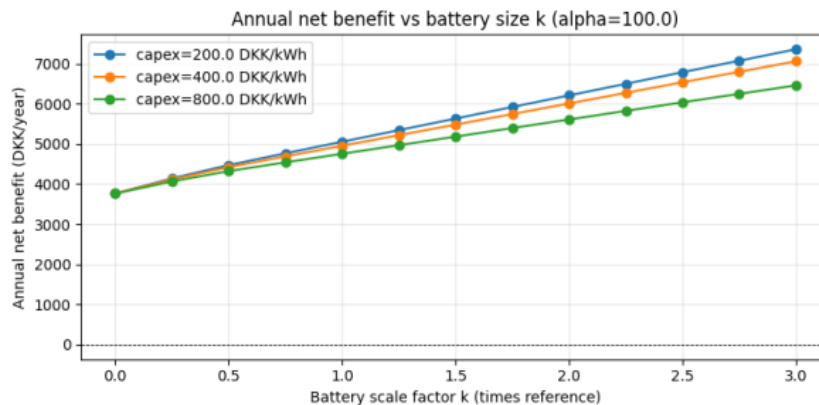
✗ Insufficient — A figure that shows an obviously wrong solution, unnoticed (2025, consumer flexibility)



Original caption: "Hourly energy flows in the Price spike scenario: PV generation, Load, Import, Export, Curtailment." The y-axis has no label.

Why. The whole daily energy requirement is bought in hour 0 and the load is zero for the other 23 hours; the accompanying text nevertheless claims that "the optimizer shifts flexible load into cheaper hours". A figure is also a debugging tool: this one shows a missing load bound and a misread of the input data (the daily requirement had been taken as 8 kWh instead of $8 \text{ h} \times 3 \text{ kW}$). **GRADER:** "This figure demonstrates exactly what I mentioned in the first question. Your model imports the whole daily demand in the first hour."

✗ Insufficient — Caption contradicts the plot; a "net benefit" that is positive without the asset (2025, battery investment)



Original caption: "Annual net benefit versus battery size for different consumer flexibility levels ($\alpha = 0.01, 1, 100$)" — the legend shows three CAPEX values instead.

Why. The caption describes a different figure; the "net benefit" is 3 800 DKK/year at zero battery size, so it is not a net benefit of the battery; and the lines are straight, so no optimum can appear although the text discusses one. The quantity plotted was a decision variable of the model, swept by hand. **GRADER:** "I do not understand how you derive this graph while your scale factor is supposed to be a variable [...] It seems you are giving it fixed values and just calculate the net benefit."

✗ Insufficient — A whole numerical analysis reduced to four numbers, and no figure in seventeen pages (2025, consumer flexibility)

[Table: Scenario / Optimal objective:] Original 14.37 DKK; Flat prices 19.16 DKK; High export tariffs 14.99 DKK; Low export tariffs 13.93 DKK. [...] For the scenario with higher export tariffs, the cost [is] going down to 14.99 DKK for the day [...]

Why. No hourly profiles, no primal or dual plots, the base case never analysed — and the one sentence of interpretation misreads the table (14.99 is an increase from 14.37). **GRADER:** "you should also analyse the original case: what are the import/export patterns?"; "You make good points in this analysis but you are not using a lot of numerical results to support it."

✗ Insufficient — Comparing costs across scenarios that do not serve the same energy (2025, consumer flexibility)

The cheapest scenario is of course the one with a higher discomfort, but this one does also not satisfy the same load requirements as the others do. The load consumed is only at about 3.945 kWh while the other scenarios are at 27.94 kWh.

Why. A cost of -12.57 DKK for 3.9 kWh is not comparable with 28 DKK for 27.9 kWh; the comparison needs a common basis (cost per kWh served, or cost at equal discomfort). Recognizing that the metric is wrong and then still using it is not an analysis.

✗ Insufficient — Scenario values chosen without reference to the other parameters; solver noise reported as results (2025, consumer flexibility)

[Table:] $\alpha_{small} 0.01$ — objective -0.9501 — deviation 1.44×10^{-11} ; $\alpha_{medium} 1$ — objective -0.9501 — deviation 7.84×10^{-10} ; $\alpha_{large} 100$ — objective -1984.93.

Why. A discomfort weight of 0.01, 1 or 100 DKK/kWh² means nothing unless it is compared with the prices (0.85–2.5 DKK/kWh): at 100 the load is certain to follow the reference exactly, at 0.01 the weight is irrelevant, so two of the three scenarios test nothing. The identical objectives and the 10^{-11} kWh deviations should have been read as "the model is not doing what we think". **GRADER:** "doesn't make sense to conduct such a high modification in alpha [...] The relative modifications should always be related to the rest of the parameters (electricity price and grid tariffs)."

✗ Insufficient — A table that mislabels a nodal quantity and its unit (2023, DC-OPF)

[Table 2, per generator G_1 – G_{12} and W_1 – W_6 :] Cost [\$/MWh] · Production [MWh] · Production cost [\$] · LMP [\$] · Revenue [\$] · Profit [\$]

Why. The LMP is a price per MWh, not \$; and it is a property of a *node*, so tabulating it per generator misleads the reader about how many distinct prices there are. **GRADER:** "This table is misleading for the LMPs, as the set of generators does not correspond directly to the set of nodes. –1p". In this year's assignment the analogue is the hourly balance dual λ_t : one value per hour, in DKK/kWh, never averaged over the day without saying so.

8. What the graders wrote most often

Across nine reports and three years, the remarks below account for most of the points lost. Each is a five-minute check before submission.

- Running indices and sets.** Every constraint carries " $\forall t \in \mathcal{T}$ " (and every other index it runs over); every set and every symbol is defined before it is used; the dual of an hourly constraint has an hourly index.
- The full KKT set.** Stationarity for *every* variable, primal feasibility, dual feasibility and complementarity — with multipliers for the non-negativity bounds. Derive the stationarity conditions from the Lagrangian you wrote, with its sign convention, and check every sign against it.
- The dual problem is written, then explained** (which primal variable gives which dual constraint, and why it is an equality or an inequality), not replaced by the KKT conditions.
- Extensions keep the standard of the first model.** New parameters defined with meaning, value and unit; new equations described one by one; unchanged equations referenced, not retyped.
- Convexity is argued, not asserted** ("linear objective and constraints", "binary variables, hence non-convex", "quadratic term with positive coefficient"); a Hessian of a linear function is not an argument.
- Every dual variable gets an economic sentence:** what one more unit of the right-hand side is worth, and in which hours or cases it is zero.
- Read the data files carefully** (units, what a ratio is a ratio of, what a "minimum" is a minimum of) and show the input data once.
- Sanity-check before interpreting.** Identical objectives for different parameters, 10^{-11} kWh, profits from a cost minimization, a battery paid back in three months, all energy in one hour: each is a bug, not a finding. Compare closed forms with the solver.
- Figures with units, labels, a readable legend, and a message.** Fewer series, shared axes for comparisons, and a caption that says what to look at; bulk tables in the appendix, but the argument in the text.
- Scenario values relative to the data.** A weight, a tariff or a budget is chosen in relation to the prices it competes with, one parameter at a time, with the purpose stated before the numbers.
- Text and numbers from the same run.** Regenerate every figure and table from the code before writing the final version; describe in the report which script produces what.

12. **Write the answer, not the process.** No "it should have been shown that...", no debugging diary, no group-process remarks; state what was done, what it shows, and what it means.

9. What the best reports had in common

They used the same template for every model — parameters, decision variables, objective, constraints each named and described, properties (class, convexity, strong duality), a multiplier table, the Lagrangian, the KKT conditions in four blocks, the dual, its interpretation — so that later questions could be stated as "the model of Question 1 with ... replaced by ...". Every symbol was defined once, with a unit and a role, and new symbols were introduced in a dedicated block when a model was extended. Derivations were shown as a method (regroup the Lagrangian per variable, read off dual feasibility, keep the constant part; write complementary slackness for every variable and turn it into if–then rules), not as a pasted result. Modelling choices were motivated in prose, and invented parameter values were anchored in a source or in the magnitudes of the data. Results sections mirrored the question, with fixed paragraph headings (primal results, dual interpretation, effect of the cost structure, comparison with the qualitative analysis, conclusion), and every experiment was announced with its purpose, its scenarios and its metrics before any number appeared. Tables mixed primal totals, activity counts and dual values across scenarios; figures were multi-panel with a shared time axis, bars for energy and lines for prices or duals, one legend and a caption stating the message. Theory and numbers were cross-checked explicitly — a bound binds if and only if its multiplier is positive; a flexible asset is idle when the price signal gives it nothing to do — and anomalies were discussed rather than hidden. Finally, they were honest about scope: a limitations paragraph that says which way each assumption biases the result, a negative result reported with its diagnosis, and a declaration of the AI tools used and of how their output was checked.